

Sunanda Samanta

Software Engineer — Full-Stack & Backend

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SUMMARY

Software engineer with ~4 years building scalable web applications and backend services end-to-end. Strong across the MERN stack and Node.js/TypeScript, with working Go and Python, and hands-on with microservices, event-driven systems (Kafka), and Kubernetes-based deployment. Built real-time systems at sub-50ms latency, a production AI agent, and DevUtils — a developer-tools product I grew solo to 5,000+ monthly users. Comfortable owning features from design through deployment, on-call, and monitoring.

EXPERIENCE

Software Engineer | Perpetuum Innovations Pvt. Ltd — Gurugram Jul 2024 – May 2026

- Built the backend platform from an empty repo and made the early architecture calls — a modular monolithic, that the cross-platform apps shipped on. Owned it end-to-end including deployment, CI/CD, and Cloud.
- Built a real-time messaging system on Pub/Sub, held 1,000+ concurrent sessions at sub-50ms latency in prod.
- Deployed and ran services on Kubernetes with Helm and Terraform, containerized with Docker, and automated releases via GitHub Actions.
- Designed database schemas and indexes and reworked the slowest queries, cutting average API response time by about 40%.
- Built async and data-sync pipelines with retry, failure handling, and replay, and stood up monitoring and alerting for production visibility.

Full-Stack Developer | VebHolic Pvt. Ltd — Remote Sep 2021 – Jul 2024

- Built full-stack MERN features end-to-end for a SaaS product (~10,000 users): React front ends, Express/Node.js REST APIs over MongoDB, PostgreSQL, authentication, third-party integrations, and monitoring.
- Shipped a production AI agent — an LLM with retrieval (RAG over vector search) with custom Llama model deployment that let users act on internal data in plain language. Cut routine manual lookups roughly 60%.
- Wrote Python and Node.js for production APIs and data integration; reworked batch and background processing for heavier workloads.
- Improved performances, cutting page load times by about 25% using caching and micro frontend architecture.

SELECTED PROJECTS

DevUtils — Privacy-first developer tools I built and grew solo to 5,000+ monthly active users. Processes multi-GB files entirely in the browser via Service Worker chunking and streaming, no backend involved at all.

High-Frequency MEV / Arbitrage Bot (Rust) — Low-latency on-chain bot using mempool monitoring and flash loans with sub-second execution. Built to learn on-chain systems end-to-end.

Infinite-GIF — Open-source engine for optimized, continuous GIF rendering and generation. Kind of browser stenography.

SKILLS

Languages: JavaScript / TypeScript, Go, Python, Rust, SQL, HTML/CSS

Frontend: React, Next.js, Tailwind, Flutter, Angular

Backend: Node.js, Express, microservices, serverless, REST APIs, WebSockets, event-driven messaging (Kafka, Pub/Sub), Redis caching

Databases: MongoDB, PostgreSQL, Redis, Firestore, ArangoDB, schema design, indexing, query optimization

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Helm, Terraform, CI/CD (GitHub Actions), Git, Jest, Playwright

AI: production AI agents, LLM integration, RAG / vector retrieval, AI workflow automation, AI dev tools (Cursor, Copilot, Claude)

EDUCATION

B.Tech, University of Engineering & Management, Jaipur 2018 – 2022 • CGPA 8.9